

Lab. 4 – Numerical Analysis of Patch Array

Assigned: Friday May 15, 2026
Due: Friday June 5, 2026

Notes

To take the exam with the grading of submitted assignments + oral discussion it is necessary:

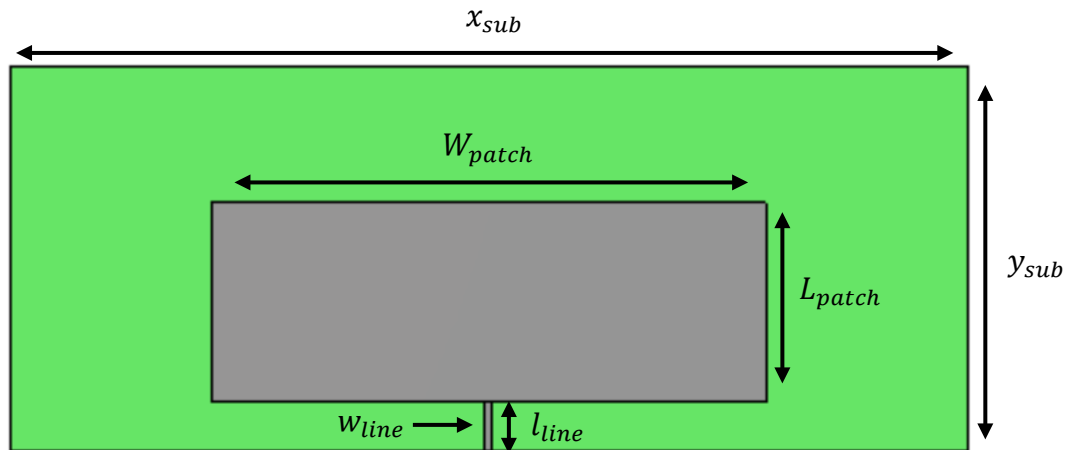
1. to prepare **individually** the report of the lab, that must include the obtained results and their brief comment;
2. to submit the report **within the deadline**, uploading the PDF in the section “Elaborati” of the web portal

The lab is focused on the numerical analysis of a patch array designed to radiate in the broadside direction. The array is printed on a FR4 substrate, with relative dielectric constant $\epsilon_r = 4.45$ without losses ($\tan\delta = 0$) and thickness $h = 1.55\text{mm}$. The design frequency is $f_0 = 5.8\text{ GHz}$.

PART 1 – Analysis of the radiating elements

A. Analyze a single patch fed by a microstrip line using CST Studio Suite, as shown below.

Plot the magnitude of the input scattering parameter over the frequency band 5.3–6.3 GHz. Plot the gain in the E and H planes and evaluate its maximum.



Geometrical parameters:

$$x_{sub} = 50\text{ mm}$$

$$y_{sub} = 20\text{ mm}$$

$$W_{patch} = 29\text{ mm}$$

$$L_{patch} = 10.4\text{ mm}$$

$$w_{line} = 0.39\text{ mm}$$

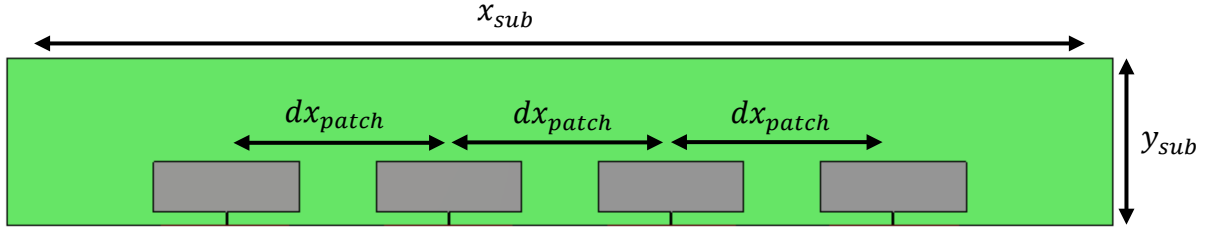
$$l_{line} = 2.5\text{ mm}$$

B. Consider now the patch embedded in the array layout.

Plot the magnitude of the input scattering parameters (self-terms only) over the same frequency range. Are they identical? Compare these results with those obtained in Part 1A. Do they still match?

Plot the gain of the individual patches in both the E-plane and H-plane. Are they identical? Determine the maximum gain and compare it with the corresponding values from PART 1A.

Finally, combine the fields (assuming equal amplitude and phase) to obtain a preliminary estimate of the radiation pattern of the full array.



Geometrical parameters:

$$x_{sub} = 220 \text{ mm}$$

$$y_{sub} = 33.1 \text{ mm}$$

$$W_{patch} = 29 \text{ mm}$$

$$L_{patch} = 10.4 \text{ mm}$$

$$dx_{patch} = 44.2 \text{ mm}$$

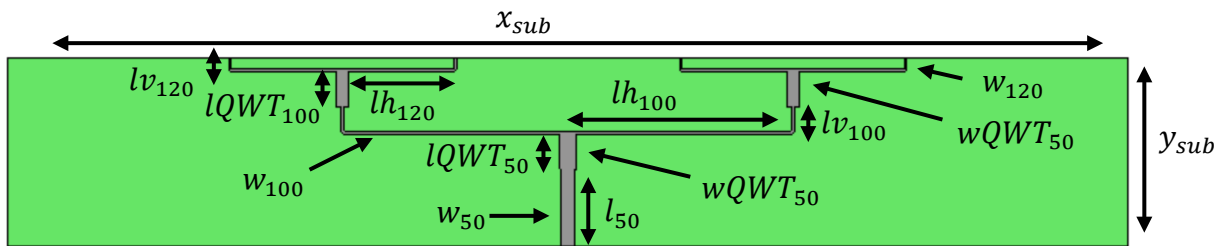
$$w_{line} = 0.39 \text{ mm}$$

$$\ell_{line} = 2.5 \text{ m}$$

PART 2 – Analysis of the BFN

Using CST Studio Suite, analyze the BFN shown below. Evaluate and plot the frequency behavior of the scattering parameters (magnitude and phase).

To speed up the analysis, please feed only the 50Ω microstrip line.



Geometrical parameters:

$$x_{sub} = 220 \text{ mm}$$

$$y_{sub} = 36.9 \text{ mm}$$

$$lh_{100} = 44.2 \text{ mm}$$

$$lh_{120} = 22.1 \text{ mm}$$

$$w_{50} = 2.94 \text{ mm}$$

$$wQWT_{50} = 3.1 \text{ mm}$$

$$lv_{100} = 5 \text{ mm}$$

$$lv_{120} = 5 \text{ mm}$$

$$wQWT_{100} = 2.4 \text{ mm}$$

$$l_{50} = 15 \text{ mm}$$

$$\ell QWT_{50} = 7.2$$

$$w_{100} = 0.68 \text{ mm}$$

$$w_{120} = 0.39 \text{ mm}$$

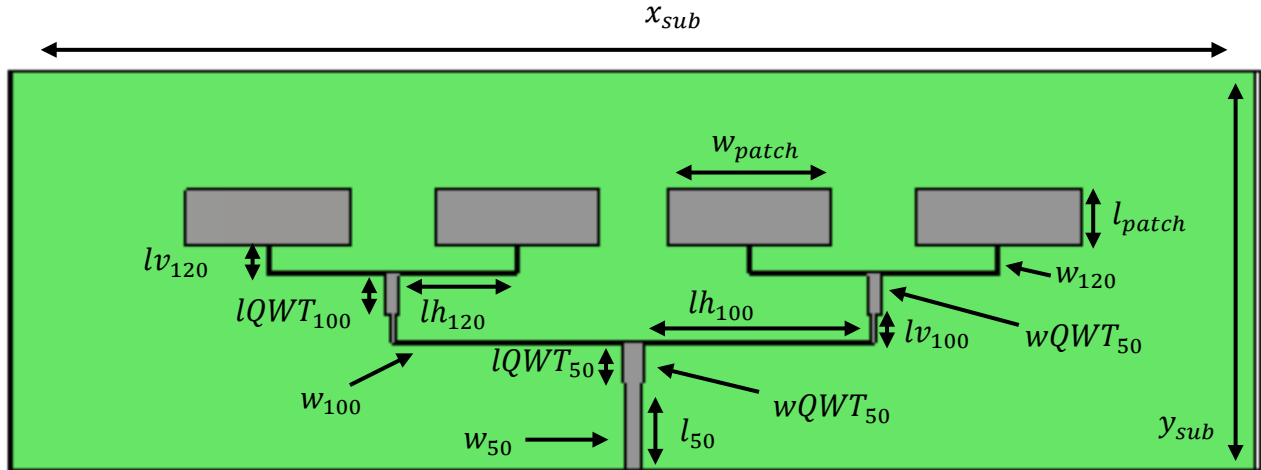
$$\ell QWT_{100} = 7.2$$

PART 3 – Analysis of the full array

Using CST Studio Suite, simulate the entire array.

Plot the magnitude of the input scattering parameter over the frequency band 5.3–6.3 GHz.

Plot the gain of the array in both the E-plane and H-plane. Determine the polarization along the direction of maximum radiation and the maximum gain. Compare the radiation pattern with the one “artificially” obtained in PART 1B.



Geometrical parameters:

$x_{sub} = 220 \text{ mm}$	$w_{50} = 2.94 \text{ mm}$	$l_{50} = 15 \text{ mm}$
$y_{sub} = 70 \text{ mm}$	$w_{QWT_{50}} = 3.1 \text{ mm}$	$l_{QWT_{50}} = 7.2$
$lh_{100} = 44.2 \text{ mm}$	$lv_{100} = 5 \text{ mm}$	$w_{100} = 0.68 \text{ mm}$
$lh_{120} = 22.1 \text{ mm}$	$lv_{120} = 5 \text{ mm}$	$w_{120} = 0.39 \text{ mm}$
	$w_{QWT_{100}} = 2.4 \text{ mm}$	$l_{QWT_{100}} = 7.2$
	$l_{patch} = 10 \text{ mm}$	$w_{patch} = 29 \text{ mm}$

PART 4 – Measurements

Using the VNA (after its calibration), measure the variation of $|S_{11}|$ of the array prototype over the same frequency range considered in PART 3.

Compare the results obtained by the experimental characterization with those deriving from its numerical simulation. Represent the frequency behavior of $|S_{11}|$ (in dB) on the same plot and comment the obtained results.

Compare E and H planes measurements in anechoic chamber with CST simulations.